IMAGE SEGMENTATION PROJECT

# AI VISION EXTRACT

## Automated Subject Isolation using Deep Learning

Semantic Image Segmentation with DeepLabV3

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# PROJECT OBJECTIVE

The primary goal is to build a robust Machine Learning model capable of automatic subject extraction from images. By utilizing Deep Learning, we aim to achieve high-precision semantic segmentation.

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### CORE CHALLENGE

Achieving accurate pixel-wise classification to seamlessly separate complex foreground subjects from diverse backgrounds.

## Key Specifications

- **Input:** Any RGB image containing a discernible subject and background context.
- **Output:** Subject fully isolated on a pure black or transparent background.
- **Applications:** Photography automation, virtual conferencing, AR/VR integration, and background replacement.
- **Method:** DeepLabV3 architecture utilizing a pre-trained ResNet101 backbone.

# LEARNING OUTCOMES

- **Semantic Segmentation:** Mastered principles of pixel-wise classification and deep learning architectures like DeepLabV3.
- **Data Pipeline:** Implemented preprocessing workflows for mask annotations and pixel-wise tasks.
- **Model Evaluation:** Utilized IoU and Dice coefficient metrics for rigorous model validation.
- **Deployment:** Built a production-ready solution with a responsive Streamlit web application.
- **MLOps Mastery:** Executed end-to-end workflow from raw data ingestion to inference API.

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# COCO 2017 DATASET ANALYSIS & PROCESSING

## DATASET OVERVIEW

Large-scale object detection & instance segmentation benchmark serving as the foundation for our model training.

### Training Set Statistics

- **118,287** total images in raw dataset
- **117,254** valid images after filtering (99.1% retention)
- **80** distinct object categories with high-quality instance masks

### Validation Set

- **5,000** total images available
- **4,952** valid after filtering (99.0%)
- Maintains same 80 categories for consistent evaluation

### Processing Pipeline

Implemented robust anomaly detection to handle missing files and corrupted images.

- Minimum box area threshold: **32 pixels** (5.7×5.7)

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# IMAGE DATA PREPROCESSING

## ANOMALY DETECTION PIPELINE

Automated identification of missing files and corrupted images. Filters invalid annotations and eliminates tiny bounding boxes below the 32-pixel threshold to ensure data integrity.

## MASK EXTRACTION PROCESS

- Converts instance masks to semantic masks with COCO category IDs (0-80).
- Merges overlapping masks using RLE encoding.
- Saves processed masks as lossless PNG files.

## QUALITY ASSURANCE

Achieved >99% valid data retention rate with detailed anomaly reports and binary validation checks for every image in the dataset.

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Part 2/2: Image Transformation Pipeline

# MODEL INPUT NORMALIZATION

- **Resize Operation**:
  Target size of 520×520 pixels to meet ResNet101 backbone requirements while preserving aspect ratio considerations.
- **Tensor Conversion**:
  Images are normalized to range [0, 1] and undergo channel-wise normalization using standard ImageNet statistics.
- **Batch Processing**:
  Implemented with a batch size of 8 images per iteration, GPU-optimized for Kaggle/Cloud environments.

| Normalization Parameters | |
|---|---|
| Mean (RGB) | [0.485, 0.456, 0.406] |
| Std (RGB) | [0.229, 0.224, 0.225] |

**Rationale**:
The pre-trained ResNet backbone expects inputs normalized with specific ImageNet statistics. Failing to match this distribution would significantly degrade transfer learning performance.

# DEEPLABV3 RESNET101 ARCHITECTURE

- **Backbone: ResNet101** — A 101-layer residual network pre-trained on ImageNet. Extracts multi-scale features with strides of 4, 8, and 16.
- **ASPP Module** — Atrous Spatial Pyramid Pooling captures multi-scale contextual information using dilated convolutions at different rates.
- **Decoder** — Upsamples feature maps to original resolution. Outputs 81 channels (80 COCO categories + 1 background) for pixel-wise classification.
- **Loss Function** — Utilizes CrossEntropyLoss to handle multi-class segmentation and class imbalance across the dataset.

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# TRAINING CONFIGURATION

### Optimization Strategy

**Optimizer:** SGD (Stochastic Gradient Descent)
**Learning Rate:** **0.01** with Momentum **0.9**
**Regularization:** Weight decay **1e-4** (L2) to prevent overfitting

### Batch & Hardware

**Batch Size:** **8 images** per iteration
**Environment:** GPU-optimized for Kaggle/Colab (NVIDIA)
**Data Handling:** Shuffled training loader, sequential validation

### Training Loop

**Forward:** DeepLabV3 outputs dictionary with 'out' key
**Loss:** CrossEntropyLoss vs. ground-truth masks
**Backward:** Loss tracking & weight updates via SGD

### Inference Mode

**Configuration:** Model set to **eval()**
**Optimization:** **torch.no_grad()** disables gradients
**Stability:** Batch Norm uses running statistics; no dropout

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# STREAMLIT WEB APPLICATION

INTERACTIVE DEPLOYMENT PLATFORM

### USER INTERFACE

Responsive dark-mode design with sidebar settings panel for real-time control adjustments.

### IMAGE PROCESSING

Support for single or batch uploads (JPG/PNG). Real-time inference with side-by-side comparison.

### CUSTOM BACKGROUNDS

Instant replacement with Pure Black (#000), White (#FFF), or Studio Gray (#2D2D2D).

### ENHANCEMENTS

Post-segmentation controls for Brightness (0.5–2.0×) and Contrast adjustment.

### ADVANCED FEATURES

Auto-crop to subject, subject coverage metrics (%), and original resolution display.

### EXPORT OPTIONS

Download processed images as PNG or batch ZIP with full metadata tags included.

# PROJECT OUTPUT & FUTURE SCOPE

## Current Achievements

- **High Performance Metrics:** Tracked via IoU, Dice Coefficient, and Pixel-wise Accuracy.
- **Robust Dataset Processing:** 122,206 clean semantic masks extracted with 99%+ quality retention.
- **Production Deployment:** GPU-optimized Streamlit application capable of batch processing.

## Future Roadmap

- **Domain-Specific Fine-tuning:** Optimization for e-commerce products, portrait photography, and medical imaging.
- **Real-Time Video Processing:** Frame-wise segmentation with temporal smoothing for live background replacement.
- **Advanced Model Optimization:** ONNX export, INT8 quantization, and distillation for edge devices.